

Arjun Nanduri

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Education

University of California: Berkeley, BA in Computer Science

Expected May 2027

- GPA: 3.9
- SAT: 1590
- Courses: Probability & Stochastic Processes, Machine Learning, Computer Security, Data Structures

Research

Data Science Research Intern, CDSS, UC Berkeley – Berkeley, CA

Jan 2025 – May 2025

- Fine-tuned BART transformer for Ancient Akkadian lemmatization in low-resource settings, beating open-source baselines by 15% using a custom edit-distance metric over lemma forms.
- Built preprocessing and alignment pipelines for structured linguistic data from ORACC's cuneiform corpus (~1M annotated words across ~10K texts).
- Stepped up to lead disorganized team with no formal authority — built shared task system, presented at UC Berkeley Discovery Symposium.

Experience

Machine Learning Engineer, Aver Technologies Inc. – Virginia

Aug 2025 – Present

- Built multi-stage PyTorch CV pipeline on 1000fps pile-driving footage — YOLO detection and segmentation to track painted depth markings, audio-visual synchronization to isolate hammer strikes, and depth-map-based pixel-to-real-world conversion — automating a manual DOT pile depth submission process.
- Conducted field interviews with construction workers to surface real-world constraints — inconsistent hand-drawn markings, Spanish-speaking workforce — directly informing model design decisions.
- Designing two-phase roadmap: Phase 1 CV digitization of manual process, Phase 2 LIDAR elimination of line markings.

Software Engineering Intern, CACI International Inc. – Virginia

May 2025 – Aug 2025

- Built a full-stack web app for U.S. Customs field officers using Angular, Spring Boot, and PostgreSQL.
- Expanded database test coverage by 20%, enhancing system stability and enabling more efficient bug detection.

Software Engineering Intern, Floras – Berkeley, CA

Sep 2024 – Dec 2024

- Built a Flutter mobile app for carbon offset purchases, integrating Patch.io to display ESG data by region.

Software Engineering Intern, IT Hired Guns Inc. – Orlando, FL

Jan 2023 – Aug 2024

- Published B2C cross-platform loyalty apps (e.g., HotSpot, Kenjo) for unbranded gas stations using Flutter.

Projects

Scripted Survey Browser AI Agent, CalHacks Prize Winner

<https://devpost.com/software/potbot>

- Won sponsor prize at CalHacks — built agentic pipeline using Llama vision model to validate infrastructure damage photos and auto-submit repair requests to SF city government via Playwright browser automation.
- Led team of 3, owned backend and CV pipeline; shipped working SF demo in 36 hours over an unfinished feature set.

Quadruped Locomotion with PPO (MuJoCo, Gymnasium, SB3)

github.com/FavouritePlayer/ant_sim

- Trained PPO quadruped policies in MuJoCo/Gymnasium; terrain-adapted policy achieves 2.1x reward (893 ± 130 vs. 424 ± 106) and 40% lower fall rate than the flat-trained baseline on unseen terrain, validated across 10 matched seeds.
- Built leg-amputation-robust specialist policies (~49x flat-baseline reward) and a compound policy router that switches checkpoints by damaged leg, beating any single policy across a 4-leg $\times$ 10-seed evaluation sweep.

Skills

Languages: Python, Java, JavaScript, SQL, Dart, Scheme

Technologies: PyTorch, Transformers (BART), scikit-learn, Pandas, NLTK, spaCy, MuJoCo, Gymnasium, Stable-Baselines3, React.js, Angular, Spring Boot, Flutter, Next.js, Firebase, AWS, MongoDB, Git.